

Sebastian Sjøen-Tollaksvik

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Education

Artificial Intelligence Engineer – Master

Started: 07.2023, Graduation Date: 06.2028

UiT The Arctic University of Norway

Specializing in Machine Learning

Study Abroad

Porto, Portugal

Study abroad coursework in Faculty of Engineering (FEUP) and Faculty of Sciences (FCUP), University of Porto

09.2025 – 07.2026

Kopervik high school, Specialization in natural sciences

Kopervik, Karmøy municipality

Graduation Date: 06.2022

Job Experience

Institute of Computer Science, UiT The Arctic University of Norway

Tromsø, Tromsø municipality

Lead Curriculum Designer and Group Teacher

08.2026 – present

- Led the redesign of the project curriculum for around 120 students in Computational Programming (INF-0102), tailoring it to STEM students by building each project around an applied, data-driven problem
- Created two projects that progress from guided Python and NumPy exercises to students' own object-oriented design, ending in greedy route optimization on flight data and logistic regression on avalanche risk
- Leads weekly seminars in high- and low-level programming languages through problem-based learning, guiding STEM students in debugging and algorithmic thinking so they learn to translate theory into working code
- Assesses students' code submissions against the course's learning objectives and provides structured feedback, helping students identify errors independently and improve code quality over time

INESCTEC

Porto, Portugal

AI Engineer

02.2026 – 06.2026

- Built a natural language interface for power grid simulation at one of Europe's leading research institutes, enabling engineers without programming knowledge to control complex APIs in plain English
- Implemented per-agent RAG pipeline and two-stage LLM architecture separating tool selection from interpretation, reducing hallucination and increasing transparency for end users
- The system is now actively extended by other INESCTEC researchers, building new agents on the same architecture, without modifications to the core infrastructure

Royal Norwegian Navy, Coast Guard

Norwegian coast

Artilleryman and Firefighter

08.2022 – 08.2023

- Operated and maintained shipboard artillery and weapon systems safely
- Assisted in firefighting and smoke diving operations
- Strengthened teamwork, discipline and problem-solving skills in high-pressure maritime environments

Leadership & Activities

UiT The Arctic University of Norway

Tromsø, Tromsø municipality

Student Representative, AI Engineering Program Board

08.2026 – present

- One of two student representatives voted in to represent the entire AI Engineering degree program, with voting rights on curriculum content and quality assurance
- Participates in board work on learning environment and strategic direction for the full program, including review of program evaluations, key metrics, and study quality

UiT The Arctic University of Norway

Elected Class Representative

Tromsø, Tromsø municipality

08.2024 – 08.2025, 08.2026 – present

- Acted as a liaison between students and faculty to communicate concerns and suggestions
- Organized meetings and represented the program in academic and social discussions

Competitions & Awards

Nordic AI Cup 2026 – Survival Simulator challenge

Online

Top 10 of 112 teams overall, 2nd in Norway – sole developer

09.2026

- Single-handedly designed and built the entire solution in a four-day AI competition for university students across the five Nordic countries, open to teams of up to four
- Developed a stateful, rule-based multi-agent controller in Python (FastAPI) that coordinates foraging, predator evasion and reproduction using only partial, relative observations
- Implemented landmark-based ego-motion estimation, shared coordinate frames for food allocation, and an escape planner scoring candidate headings against predicted predator movement

Skills & Interests

Tech stack: Python, C, Git, REST API, PyTorch, TensorFlow, Stable-Baselines3, OpenCV, HuggingFace, Scikit-learn, Webots, ChromaDB, OpenAI API, Gemma (self-hosted)

Language: Norwegian (mother tongue), English (fluent)

Projects: Sim-to-Real Robustness Research in Autonomous Robot Navigation (PPO/DQN, 2M+ timesteps, 100% success rate with 90% LiDAR reduction), DeepSyntax LLM VS Code Extension, Computer Vision Ophthalmology Assistant, In-memory Document Search (C) ([GitHub page](#))

Interests: Rock climbing, skiing, hiking, cooking and travelling